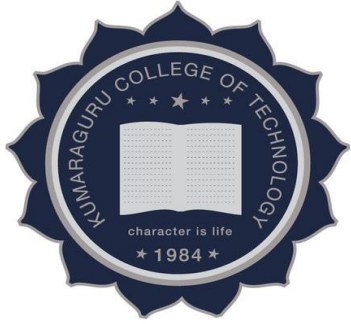




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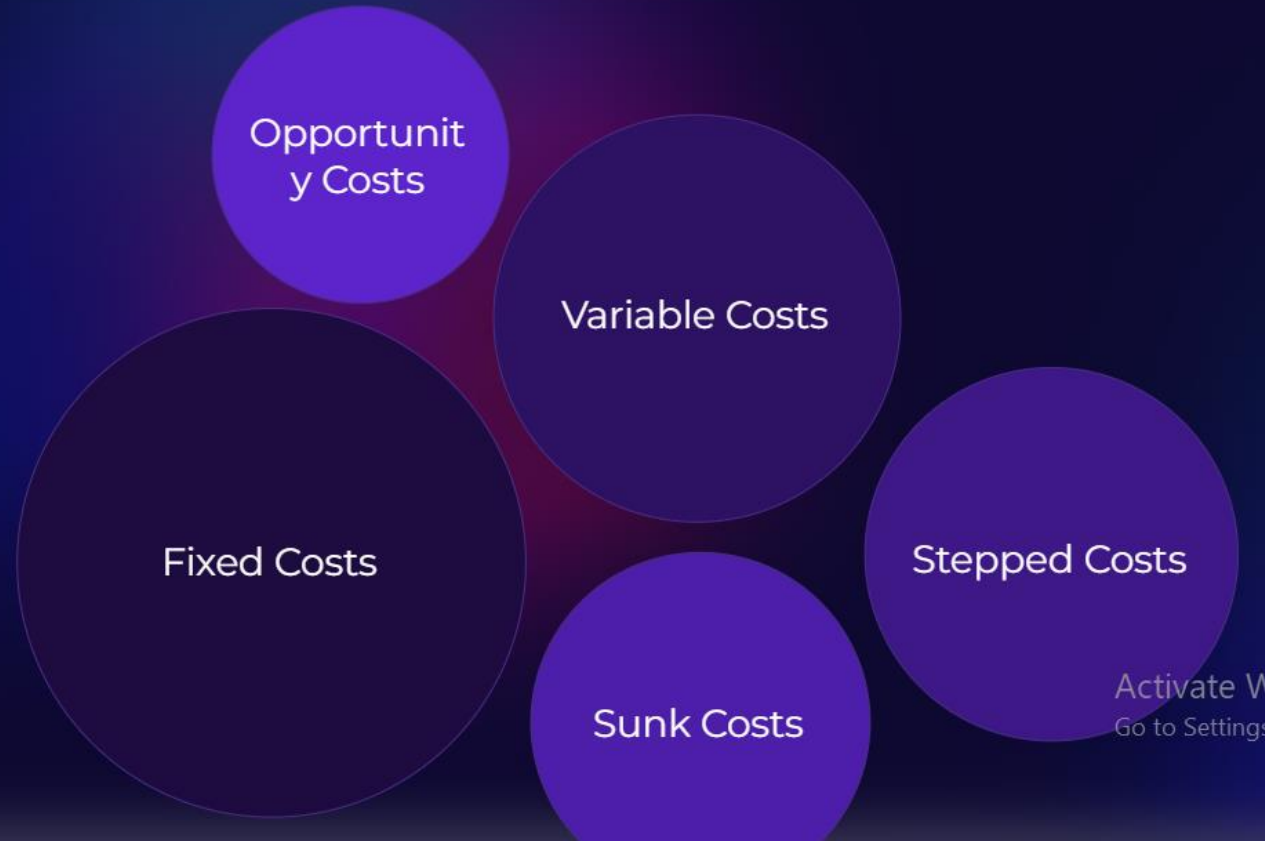
Unit -2 Cost & Financial Accounting Fundamentals

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Asst. Prof.
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Contents

- Cost concepts—fixed, variable, stepped, sunk, opportunity, relevant;
- Economies & diseconomies of scale;
- Cost-volume-profit analysis;
- Accounting principles & conventions,
- Double-entry system,
- Journal-ledger-trial balance;
- The accounting equation;
- Preparation of basic Income Statement and Balance Sheet.

Visual Summary: Cost Types at a Glance



Cost Concepts

- Fixed Cost,
- Variable Cost,
- Stepped Cost,
- Sunk Cost,
- Opportunity Cost, and
- Relevant Costs



01

Cost Concepts Overview



Fixed Cost

Costs that remain constant regardless of production volume.



❖ Fixed Cost

- Fixed costs are expenses that do not change with the level of production or sales.
- Examples include rent, salaries, and insurance.
- These costs remain constant, allowing businesses to predict their expenses more accurately over time.

Examples of fixed costs



Depreciation



Amortization



Insurance premium



Interest expense



Property tax



Rent



Salaries



Utility expenses



Legal expenses



Advertising expenses

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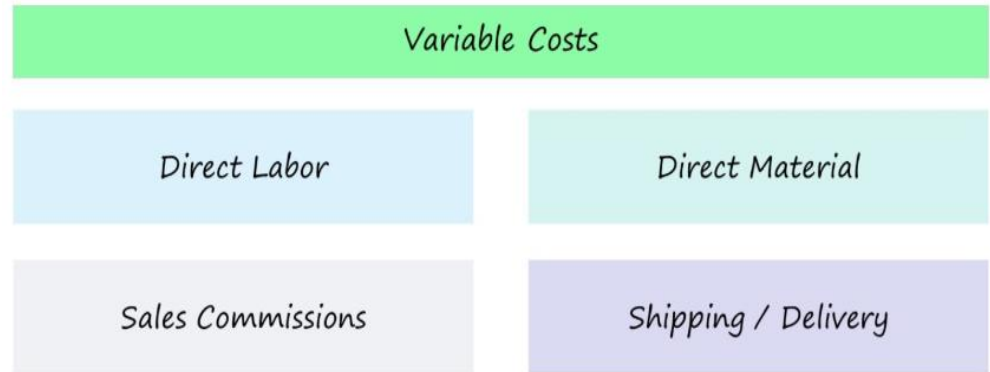
Variable Cost

Costs that vary directly with production output.



❖ Variable Cost

- Variable costs fluctuate with production levels.
- They include costs for **materials, labor, and shipping**, which increase as output rises.
- Understanding variable costs helps businesses manage budgets effectively and assess profitability **on varying production levels**.



Stepped Cost

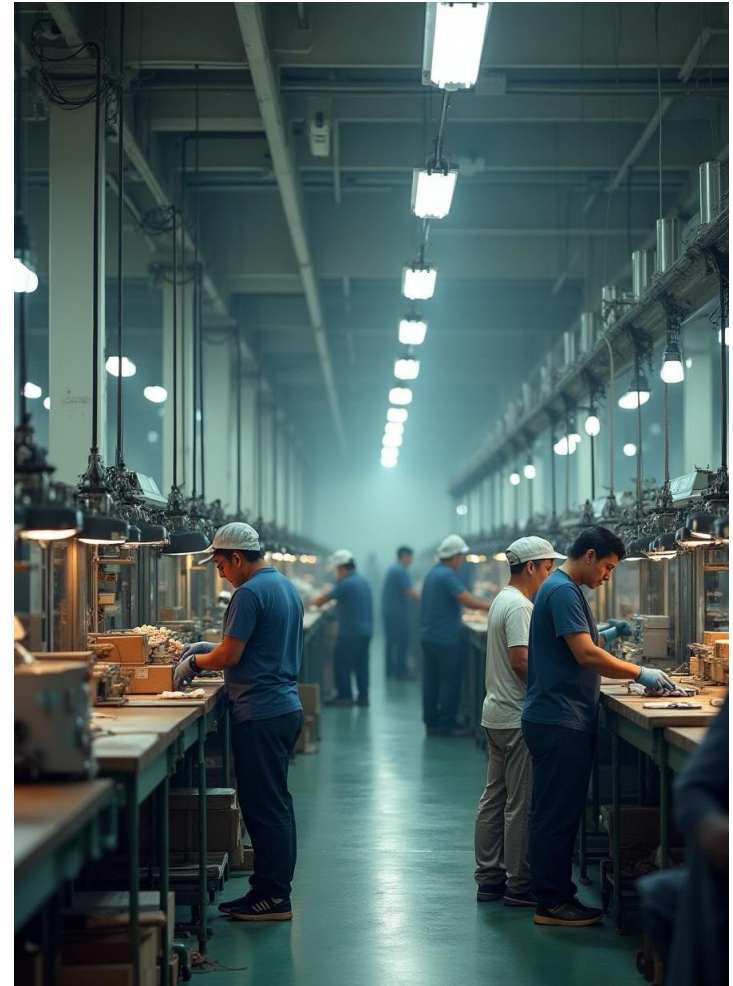
Costs that remain fixed over a range but jump to a higher level after a threshold.

- 1 Hiring an additional supervisor**
When worker count exceeds 50, requiring new management.
- 2 Leasing extra warehouse space**
Inventory exceeding current storage capacity triggers additional lease agreements.

- 3 Maintenance contracts increasing**
Maintenance fees rising after certain machine operational hours are surpassed.

❖ Stepped Cost

- Stepped costs remain fixed over certain production ranges **but increase once a threshold is crossed.**
- For example, a factory may have a fixed number of workers up to a certain output level, **but hire additional staff as production increases** beyond that point.
- This model helps manage labor costs at different production scales.



02

Decision- Making Costs



Sunk Cost

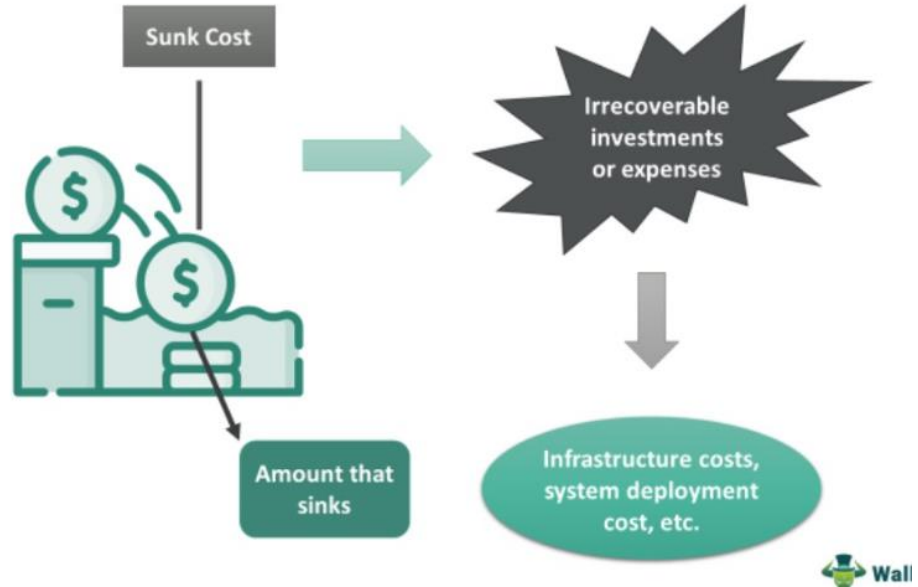
Costs already incurred and unrecoverable, irrelevant for future decisions.

- Money spent on obsolete equipment
- Non-refundable marketing campaign expenses
- Research and development costs on discontinued product
- Paid deposits on cancelled venue bookings
- Past training costs for former employees



❖ Sunk Costs

- Sunk costs are **past expenditures that cannot be recovered**.
- These costs should not influence future decisions, as they are irrelevant to future profitability.
- Businesses should focus on prospective costs and benefits when evaluating ongoing projects or investments.



Opportunity Cost

Value of the next best alternative foregone when a choice is made.

Studying vs. Working

Income lost by choosing education over immediate employment.

Project Prioritisation

Time spent on Project A, preventing work on the beneficial Project B.



Property Use

Rent foregone by using an owned building for production instead of leasing it out.

Investment Choices

Interest income lost by investing capital in machinery rather than financial assets.

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❖ Opportunity Cost

- Opportunity costs represent the potential benefits lost when one **alternative** is chosen over another.
- Recognizing these costs is vital for making informed decisions, as they highlight the trade-offs involved in allocating resources across **competing options**.



Relevant Cost

Costs that will be affected by a specific decision and should be considered.



Special Order Materials

Additional raw material cost for a unique customer order.



Overtime Labour

Incremental labour cost for production beyond regular hours.



Capacity Expansion

Cost of new equipment if increasing production capacity.



❖ Relevant Costs

- Relevant costs - directly impacted by a specific decision.
- They include **future costs that will change based on the choice being evaluated.**
- This understanding enables businesses to **make better financial decisions** by focusing on costs that truly matter in the decision-making process.



Why Understanding These Costs Matters

Smarter Budgeting & Pricing

Enables more accurate financial forecasting and competitive pricing strategies.

Improved Decision-Making

Enhances choices regarding production levels, investments, and resource allocation.

Identify Avoidable Expenses

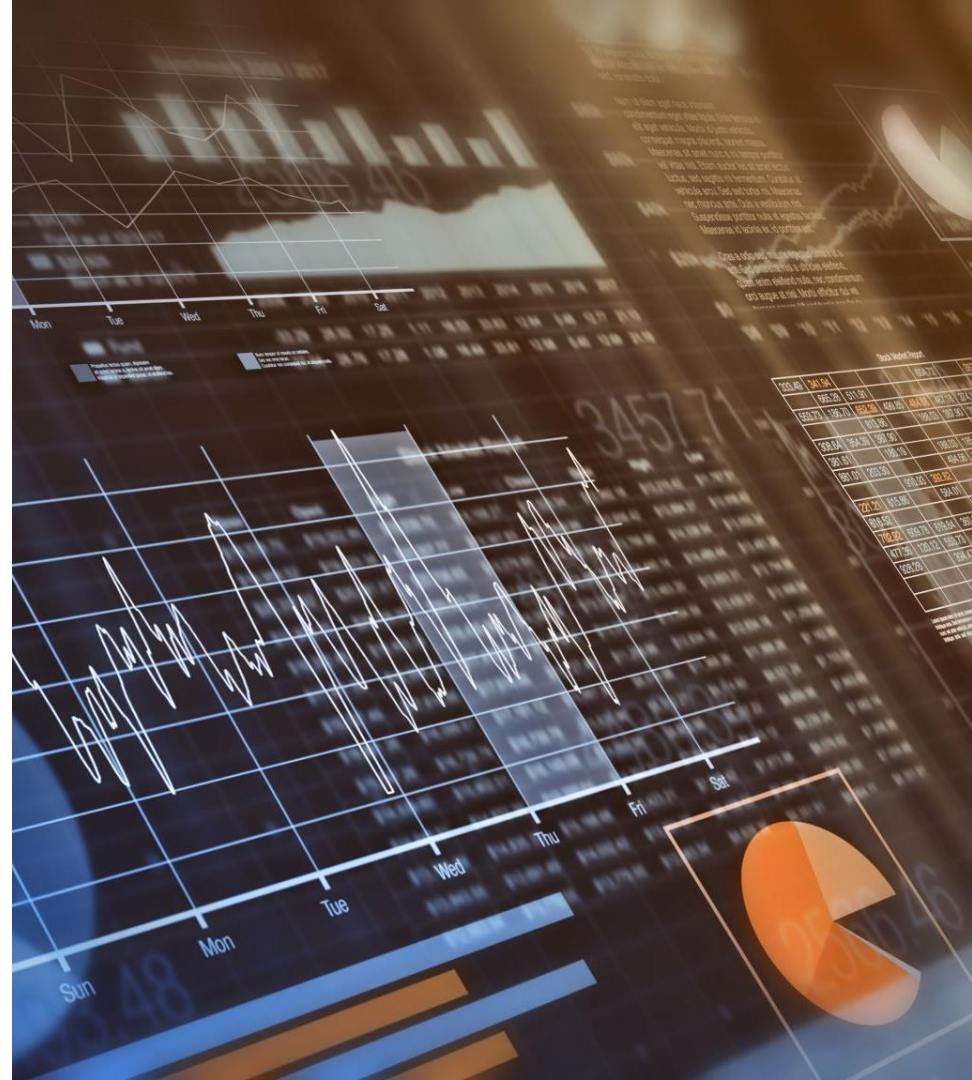
Helps distinguish between costs that can be controlled and those that cannot.

Clear Financial Impact

Clarifies the monetary consequences of various business choices.

Economics of Scale Factors

1. **Technical Economies:** Larger firms can invest in more advanced and efficient technology or machinery, which lowers the cost per unit.
2. **Managerial Economies:** Larger firms can afford specialized managerial staff, improving efficiency and productivity.
3. **Purchasing Economies:** Bigger firms can buy raw materials in bulk at discounted rates, reducing the cost per unit.



Economics of Scale Factors

4. **Financial Economies:** Large firms often have better access to capital markets and may receive lower interest rates on loans due to their size and perceived lower risk.

5. **Marketing Economies:** Large firms can spread advertising and promotional costs over a larger output, reducing the cost per unit of marketing.

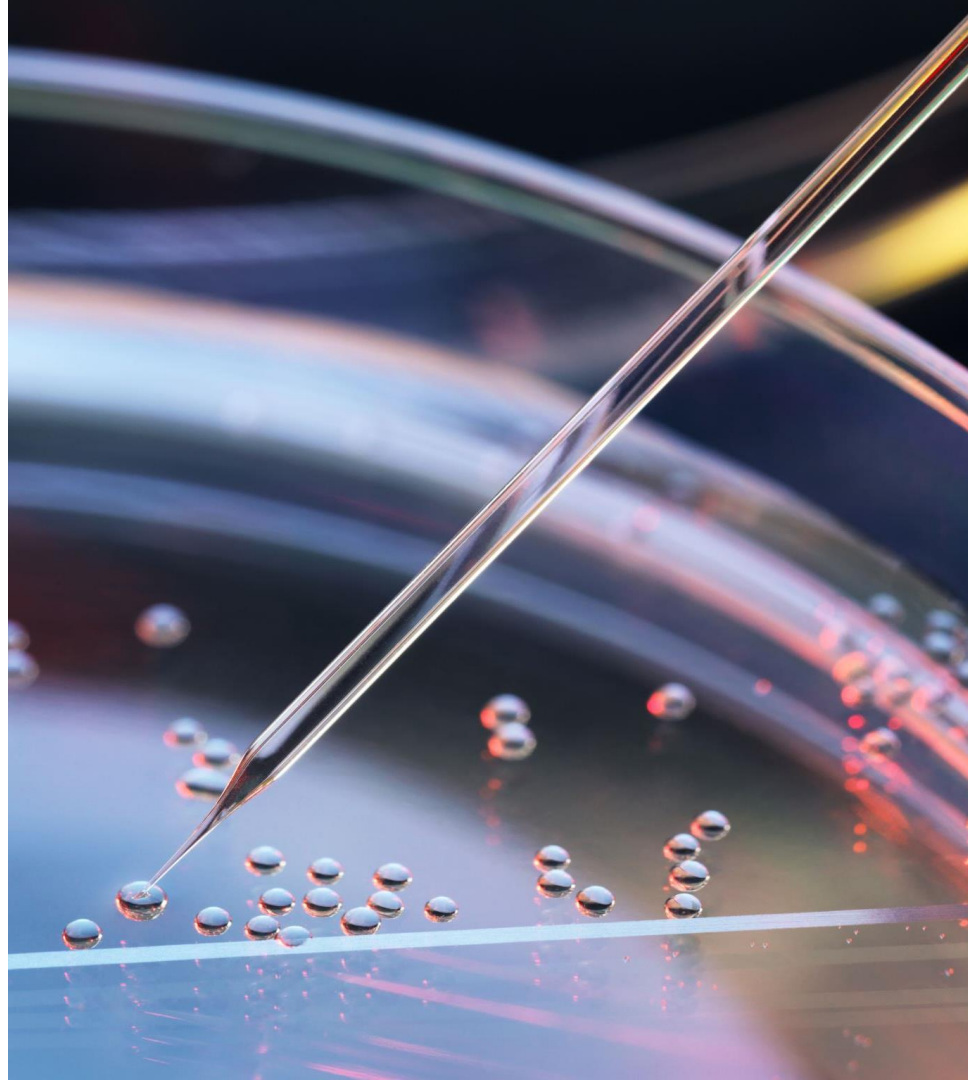
6. **Network Economies:** Some firms benefit from network effects where the value of their product or service increases as more people



125,058	154,568	95,054	124,500
125,487	56,845	97,511	125,000
124,000	110,000	99,011	154,000
105,450	150,000	99,216	95,000
86,502	35,000	101,090	154,200
	83,000	101,684	110,000
	45,000	101,962	89,000
		102,747	50,000
			68,700
			123,000

Diseconomies of Scale

- **Diseconomies of Scale** occur when a firm's average costs begin to increase as production expands beyond a certain point.



1. **Management Inefficiencies:** As firms grow larger, they may face difficulties in managing and coordinating activities efficiently, leading to increased per-unit costs.
2. **Worker Alienation:** Employees in larger firms might feel less connected to the firm's goals and less motivated, which can reduce productivity and increase costs.
3. **Bureaucratic Delays:** Larger organizations often have more layers of bureaucracy, which can slow down decision-making and reduce operational efficiency.
4. **Communication Problems:** As firms expand, maintaining effective internal communication can become more challenging, leading to inefficiencies and higher costs.
5. **Resource Constraints:** Larger firms might experience constraints in the availability of resources or face higher costs due to increased competition for inputs.

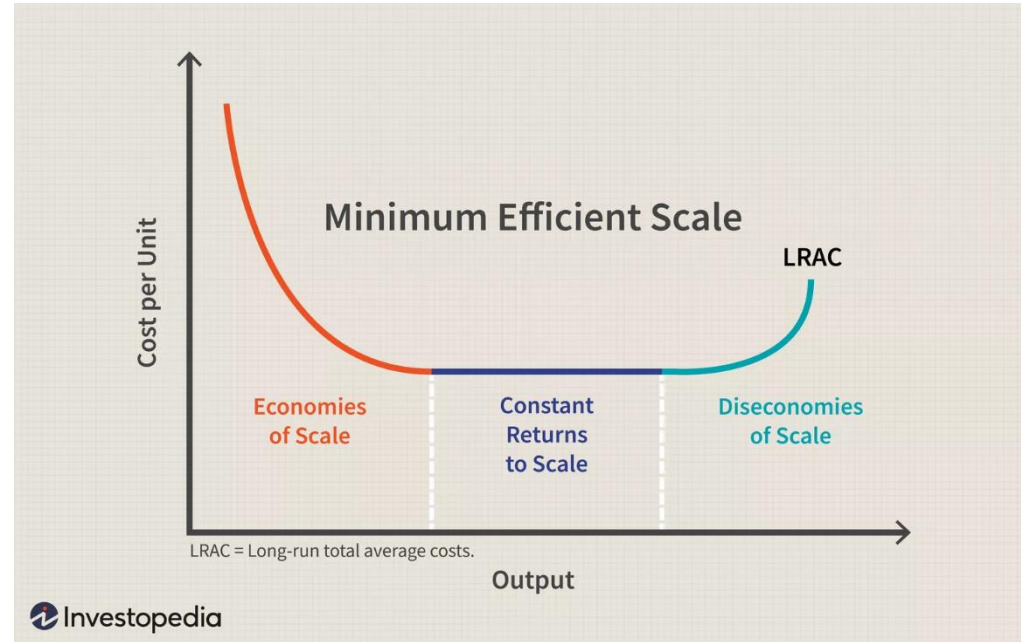
Curve

- **Economies of Scale:**

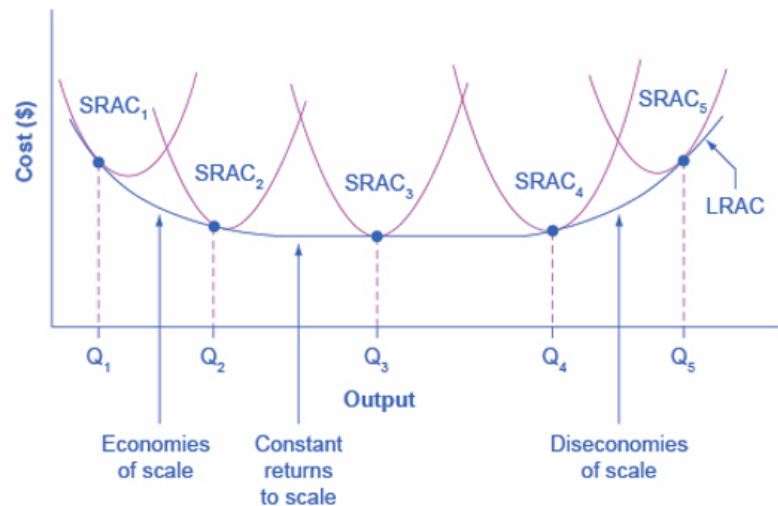
Typically depicted as a downward-sloping portion of the Long-Run Average Cost (LRAC) curve. Initially, as output increases, the LRAC decreases due to the factors mentioned.

- **Diseconomies of Scale:**

Represented by an upward-sloping portion of the LRAC curve. After reaching a certain output level, additional increases in production lead to higher average costs



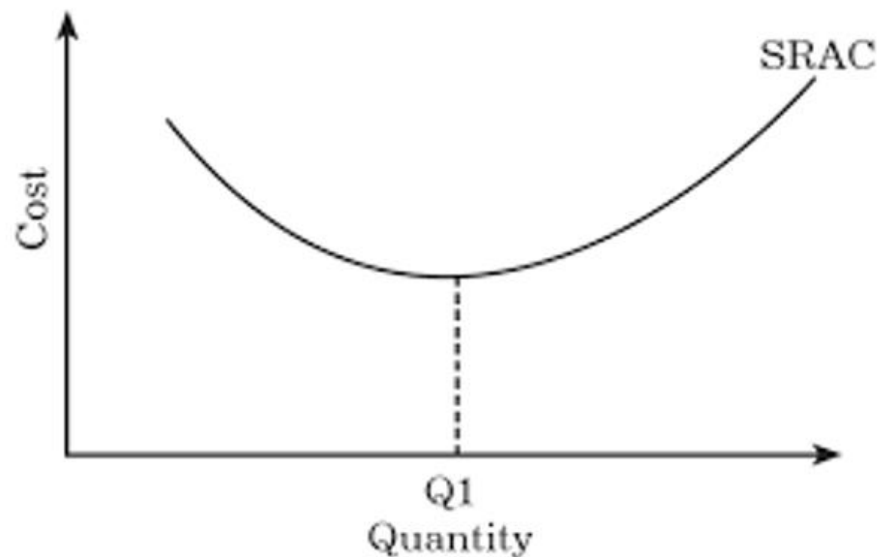
Long-Run Average Cost (LRAC):



- The LRAC curve is derived from the envelopes of all possible short-run average cost curves.
- It represents the lowest possible cost of producing any given level of output when the firm can adjust all inputs.
- The LRAC typically has a U-shape due to economies of scale (decreasing costs with increased output) and diseconomies of scale (increasing costs after a certain point).

Short-Run Average Cost (SRAC)

- The short-run average cost decreases initially as output increases due to spreading fixed costs over more units.
- However, after a certain point, it starts to increase as diminishing marginal returns set in, raising variable costs more steeply.

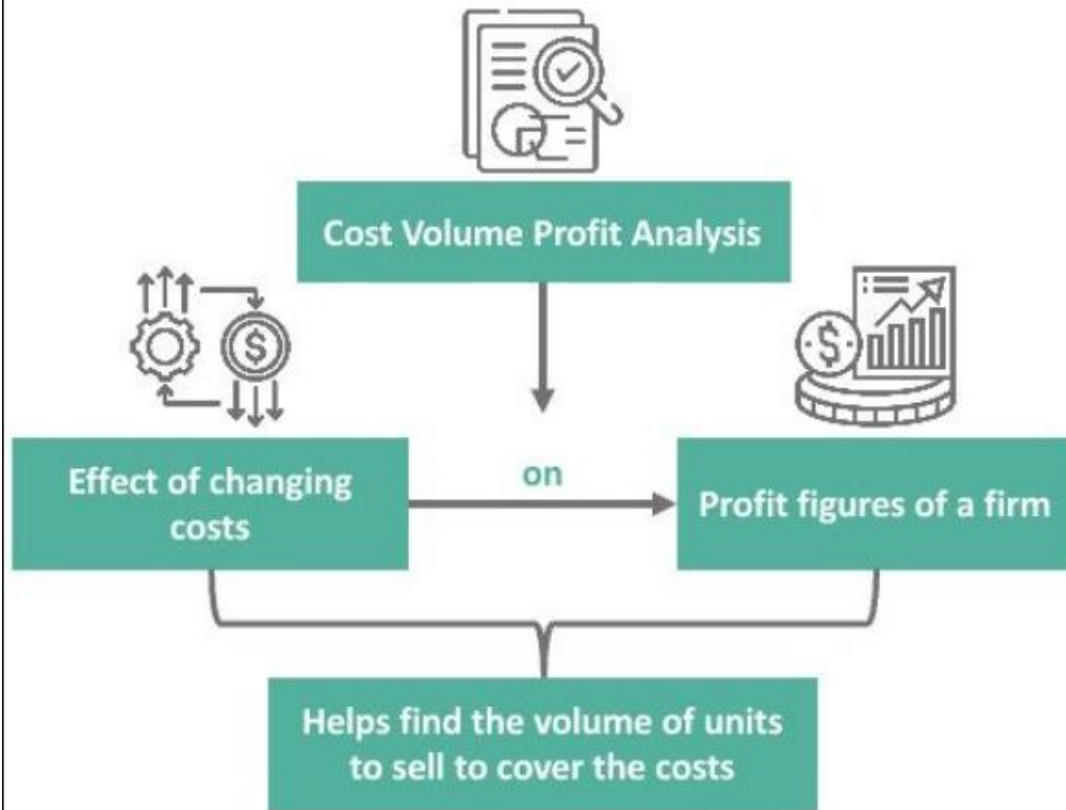




Cost-Volume-Profit (CVP) Analysis: Understanding Business Dynamics

CVP analysis is a fundamental management accounting tool designed to dissect the intricate relationships between costs, sales volume, and profit. For UK businesses, it's an indispensable guide for strategic decision-making and meticulous financial planning. By understanding how fluctuations in sales volume and costs influence overall profitability, organisations can steer towards greater financial health and stability.

What Is Cost Volume Profit Analysis?



CVP Analysis Formula For A Single Product

$$\text{Units required to achieve target profit} = Q = \frac{F + \text{Profit}}{P - V}$$

Where **F** = Total Fixed Costs

P = Selling Price Per Unit

V = Variable Cost Per Unit

P - V = Contribution Margin Per Unit

Core Concepts & Formulas for CVP

1

Contribution Margin

The revenue remaining after deducting variable costs. This figure directly contributes to covering fixed costs and generating profit. For instance, if a product sells for £50 and has a variable cost of £20, its contribution margin is £30 per unit.

2

Break-even Point

The sales volume at which total revenues precisely match total costs, resulting in zero profit. It's a crucial benchmark for viability. Calculated as:

Fixed Costs / Contribution Margin per Unit. E.g., £100,000 fixed costs / £30 contribution = 3,334 units to break even.

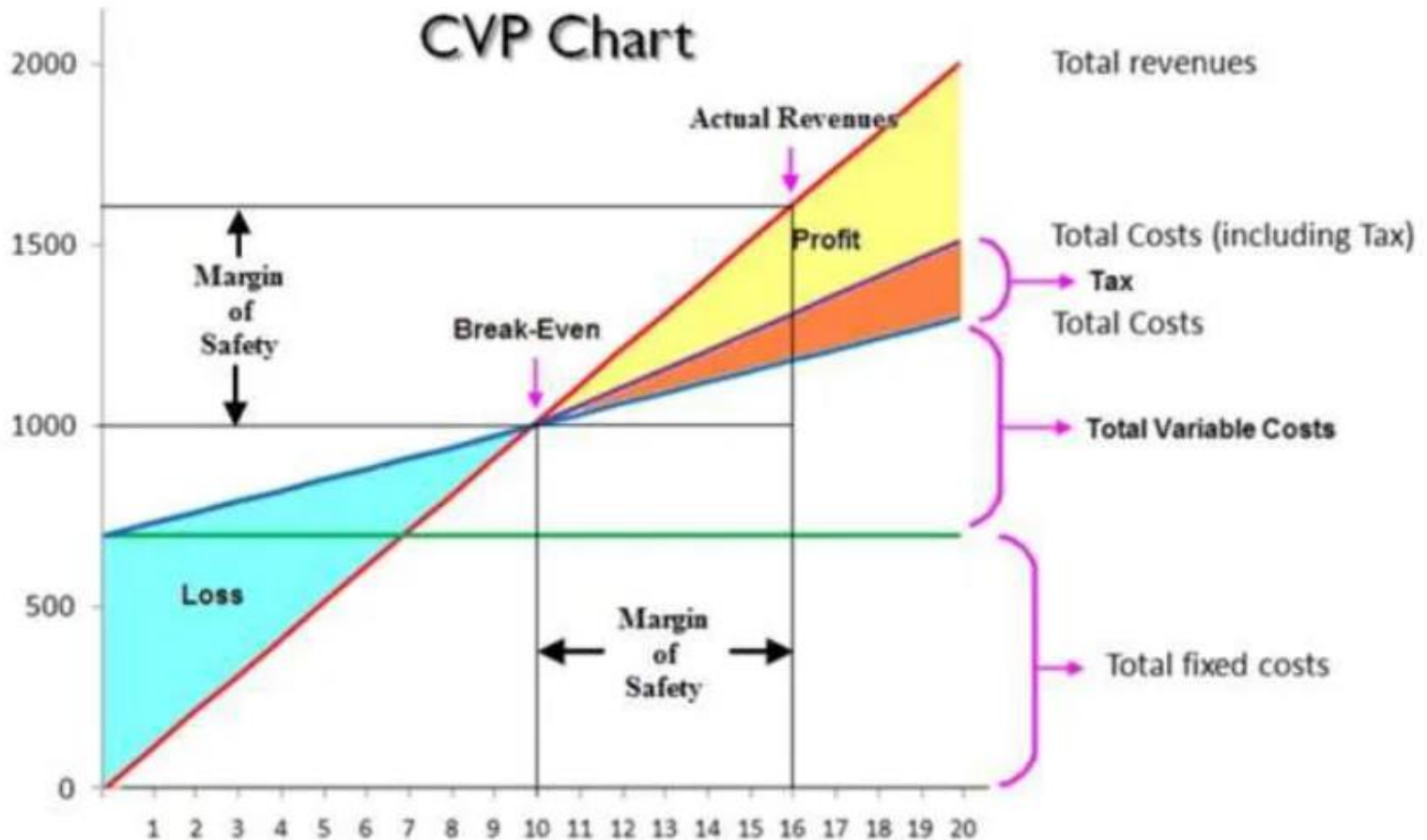
3

Target Profit Analysis

Determines the sales volume (in units or revenue) required to achieve a specific profit objective. This helps businesses set realistic goals. Formula:

(Fixed Costs + Target Profit) / Contribution Margin per Unit.

CVP Chart



Delving into the bedrock of financial integrity, this presentation explores the essential accounting principles and conventions that underpin reliable financial reporting.



Why Principles and Conventions Matter

Ensuring Accuracy and Consistency

Accounting principles are the fundamental rules guiding how financial transactions are recorded and reported, ensuring accuracy, consistency, and comparability across businesses.

Maintaining Trust and Transparency

Conventions fill gaps where formal standards don't provide clear guidance, helping accountants apply judgement while maintaining trust and transparency.

Empowering Informed Decisions

Together, these principles and conventions enable investors, managers, and regulators to make informed decisions based on reliable financial statements, fostering confidence in financial markets.

Key Principles & Conventions to Know

1 Going Concern Principle

Assumes a business will continue operating indefinitely, significantly influencing asset valuation and the preparation of financial statements. This provides a stable basis for long-term planning.

2 Accrual Principle

Records income and expenses when they occur, irrespective of when cash changes hands. This approach offers a more accurate reflection of a business's true financial performance over a period.

3 Consistency Convention

Mandates the use of the same accounting methods and procedures across different accounting periods. This ensures that financial statements are comparable over time, highlighting genuine performance trends.

4 Conservatism Convention

Encourages cautious reporting by recognising potential losses as soon as they are probable, while gains are only recorded when they are certain. This minimises overstatement of assets and income.

5 Full Disclosure & Materiality

Require that all information significant enough to influence the decisions of users should be disclosed, even if unfavourable. This ensures complete transparency and accountability in financial reporting.

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REVENUE RECOGNITION PRINCIPLE

The revenue recognition principle dictates that revenue should be recognized when it is earned and realizable, rather than when cash is received. This ensures that financial statements accurately reflect the company's performance during a specific period.

MATCHING PRINCIPLE

The matching principle states that expenses should be matched to the revenues they help generate within the same accounting period. This principle ensures that financial statements provide a clear picture of profitability during the period.

COST PRINCIPLE

The cost principle requires that assets be recorded at their historical cost, which is the amount paid at the time of acquisition. This principle provides reliability and objectivity in financial reporting, as it is based on verifiable data.

CONSISTENCY

Consistency ensures that financial statements are prepared in a uniform manner over time, allowing for accurate comparisons across periods.

PRUDENCE

Prudence requires that accountants exercise caution when making estimates and judgments, ensuring that financial statements do not overstate assets or income.

MATERIALITY

Materiality relates to the significance of financial information, guiding accountants in determining what information is essential to users of the financial statements.

MASTERING ACCOUNTING PRINCIPLES AND CONVENTIONS IS ESSENTIAL FOR CAREER SUCCESS

A solid understanding of accounting principles and conventions equips B.Tech IT students with the necessary skills to analyze financial data and make sound business decisions. This knowledge not only enhances their technical competencies but also positions them favorably in a competitive job market.





The Double-Entry System in Accounting: The Foundation of Financial Accuracy

Understanding the double-entry system is crucial for anyone engaging with financial records. This presentation will explore its core principles, historical impact, and enduring relevance in modern business.

What Is Double-Entry Accounting?

Dual Nature of Transactions

Every financial transaction is recorded twice: once as a **debit** and once as a **credit**.

The Balancing Act

This fundamental principle ensures the accounting equation remains balanced at all times: **Assets = Liabilities + Equity**.

Equal and Opposite

For every debit entry, there is an equal and opposite credit entry, maintaining financial equilibrium.

Consider this example: taking a £10,000 business loan means debiting Cash (an asset increase) and crediting Loan Payable (a liability increase). The books remain perfectly balanced.

Why Double-Entry Matters Today



Enhanced Accuracy

Provides a comprehensive, error-resistant record of all business transactions, significantly reducing discrepancies.



Fraud Detection

Helps detect errors and potential fraud by ensuring total debits always equal total credits, acting as an internal control.



Robust Financial Reporting

Essential for the preparation of key financial statements: the balance sheet, income statement, and cash flow statement.



Strategic Decision-Making

Critical for compliance with accounting standards and supports informed, data-driven business decisions.

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ACCOUNTING CYCLE



RULES OF DEBIT AND CREDIT

Personal A/c

Debit the
receiver

Credit the
giver

Real A/c

Debit what
comes in

Credit what
goes out

Nominal A/c

Debit all
expenses
and losses

Credit all
gains and
incomes

JOURNAL

- Book of original or prime entry as business transactions are recorded at the first instance.
- Transactions recorded in the order in which they occur, i.e. in **chronological order**.
- **Basic book** of accounting
- The process of entering or recording the transaction in the journal is called **Journalising**.
- Record of each transaction in the journal is called **journal entry**.

JOURNAL

Date	Particulars	LF	Debit	Credit

ADVANTAGES OF JOURNAL

- Journal provides a chronological i.e. **date wise** record of all business transactions at one place.
- Transactions recorded as and when they take place, chances of any transaction being unrecorded is reduced to the minimum.
- Facilitates preparation of ledger accounts.
- Serves as **evidence** in the court of law.
- Transactions recorded with narration, **reason** for each transaction can be ascertained.

LIMITATION OF JOURNAL

- When the number of transactions is very large, it may practically be difficult, if not impossible, to record all the transactions through one journal because of the following reasons:
 - (1) The system of recording all the transactions in a journal requires the writing down the name of the account involved as many times as the transactions occur.
 - (2) Such a system does not provide the information on prompt basis.
 - (3) Such a system does not facilitate the installation of an internal check system, since the journal can be handled only by one person.
- If we have a serious look on the limitations, it can be observed that most of them have been done away with, because of the advent of computer vis a vis accounting related softwares like tally.

LEDGER

- Book of **secondary entry**
- Transactions recorded in the journal are finally carried to the ledger.
- Contains **all the accounts** of a business whether real, personal or nominal.

DISTINCTION BETWEEN JOURNAL AND LEDGER

BASIS	JOURNAL	LEDGER
1.Nature Of Book	It is a book of prime entry	It is book of secondary entry
2.Basis Of Preparation	It is prepared on the basis of source documents.	It is prepared on the basis of journal.
3.Object	It is prepared to record all transactions in chronological order.	It is prepared to know the net effect of various transactions affecting a particular account.
4.Balancing	Journal is not balanced.	All ledger accounts are balanced
5.Narration	Narration is written for each entry	No narration is given.
6. Name of process	The process is called journalizing.	The process of recording in the ledger is called posting.

TRIAL BALANCE

"Trial balance is a statement containing the balances of all ledger accounts, as at any given date, arranged in the form of debit and credit columns placed side by side and prepared with the object of checking the arithmetical accuracy of ledger postings."

FEATURES OF TRIAL BALANCE

- It is a **statement** or a schedule.
- It contains **debit and credit balances** of various accounts.
- If the trial balance **does not agree**, it points out that there are **some errors**.
- It can be prepared only after balancing all the accounts in the ledger.
- It is prepared to check the **arithmetical accuracy** of books of accounts.

OBJECTIVES

- To provide **check on arithmetical accuracy** of books of accounts
- To provide a basis for the **preparation of final accounts**
- To provide a **summary** of ledger accounts

LIMITATIONS OF TRIAL BALANCE

- There are certain **errors** which are not disclosed by a trial balance.
- Does not provide detailed information about ledger accounts.
- It is **not a substitute** of financial statements.

The Accounting Equation

$\text{Assets} = \text{Liabilities} + \text{Owner's Equity}$



$\text{Assets} - \text{Liabilities} = \text{Owner's Equity}$

The Accounting Equation could also apply to a personal situation. Suppose you buy a car for \$5,000, borrow \$4,000 from the bank, and pay the rest yourself. Here's the result:

Accounting Equation

Assets

=

Liabilities

+

Equity



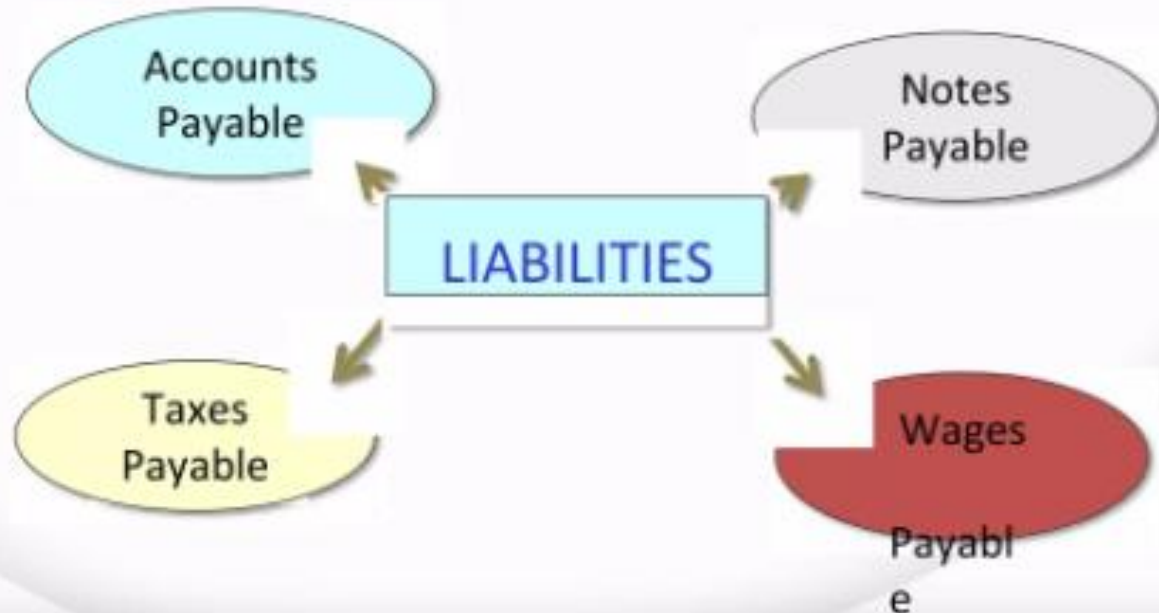
ASSETS are the **RESOURCES OWNED BY A BUSINESS**.

Here are some types of assets that might be owned by a business company:



LIABILITIES are the
CREDITOR'S CLAIMS ON ASSETS.

Creditors are the people or companies to whom a business owes something (like money).
Here are some types of liabilities that a company might owe:



EQUITY is the OWNER'S CLAIM ON ASSETS

In a business EQUITY is composed of four parts that either increase or decrease equity:



Sometimes we expand the Accounting Equation to show all the Equity components. This is called the **EXPANDED ACCOUNTING EQUATION**.



$$\text{Assets} = \text{Liabilities} + \overbrace{\text{Owner, Capital} - \text{Owner, Withdrawals} + \text{Revenues} - \text{Expenses}}^{\text{Equity}}$$



This equation must **ALWAYS BE IN BALANCE**

What is an Income Statement?

The Income Statement summarizes a company's revenues and expenses over a specific period, showing the net profit or loss. It is crucial for assessing a company's operational efficiency and profitability.

1

Key Components

Includes revenues, COGS, operating expenses, and non-operating income/expenses.

2

Financial Formula

$$\text{Net Income} = \text{Total Revenues} - \text{Total Expenses}.$$

3

Importance

Vital for assessing operational efficiency and profitability.

Structure of the Income Statement

Header Information

Includes the company name, statement title, and the reporting period.

1

2

Revenue Section

Lists total sales or service revenues generated.

Net Income Calculation

The final line details the net income or loss.

4

3

Expense Section

Breaks down costs such as COGS, operating expenses, and other expenses.

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Preparing the Income Statement

The income statement is a crucial financial document that outlines a company's revenues, costs, and expenses during a specific period. Follow these key steps to ensure accuracy in preparing the statement.

1

Gather Revenue Data

Ensure all income streams are accounted for.

2

Calculate COGS

Use the formula: Beginning Inventory + Purchases - Ending Inventory.

3

List Operating Expenses

Include salaries, rent, utilities, and marketing costs.

4

Summarize Non-operating Income

Include items like interest income and one-time gains/losses.

Understanding the Balance Sheet



Definition

The Balance Sheet provides a snapshot of a company's assets, liabilities, and shareholders' equity at a specific point in time.



Accounting Equation

It follows the accounting equation: $\text{Assets} = \text{Liabilities} + \text{Shareholders' Equity}$.



Stakeholder Assessment

The Balance Sheet helps stakeholders assess liquidity, financial stability, and capital structure.

Structure of the Balance Sheet

The balance sheet is a financial statement that provides a snapshot of a company's financial position at a specific point in time. It comprises header information, assets, liabilities, and shareholders' equity, depicting the company's financial stability and resource allocation.

Header Information

Includes company name, statement title, and date.

Assets Section

Divided into current and non-current assets for liquidity.

Liabilities Section

Shows current and long-term obligations of the company.

Shareholders' Equity

Represents owners' claims after liabilities are settled.



Preparing the Balance Sheet

1

Current Assets

List cash, accounts receivable, and inventory.

2

Non-Current Assets

Include property, equipment, and intangible assets.

3

Current Liabilities

Compile accounts payable and short-term debts.

4

Equity Section

Sum contributed capital and retained earnings.

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Importance of Financial Statements

Financial statements serve as crucial resources that help stakeholders, including investors, creditors, and regulatory authorities, to assess a company's performance and stability. They are vital for making informed decisions in the business arena.

Informed Business Decisions

Provide insights on profitability, financial health, and operational efficiency.

Investment Evaluation

Investors use these statements to assess investment opportunities and company risks.

Creditworthiness Assessment

Creditors examine for financial stability before extending loans or credit lines.

Regulatory Compliance

Authorities review financial statements for adherence to reporting requirements.